

A Domain Decomposition Approach for Finite Element Approximation of Nonlocal Diffusion Models

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FEM Course Project

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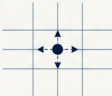
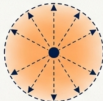
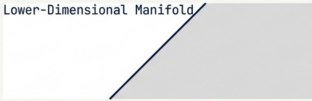
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Introduction and Problem Setting

- **Motivation:** Classical PDE-based FEM models local diffusion, but many physical phenomena involve **finite-range (nonlocal) interactions**.
- **Nonlocal diffusion models** are derivative-free, allow interactions within a finite horizon δ , and naturally admit **discontinuous solutions**.
- **Limitations of classical FEM and DD:** PDE-based DD assumes interface-only coupling, while nonlocal problems require **interior cross-subdomain interactions**, leading to denser and ill-conditioned systems.
- **Objective and scope:** Study FEM-based nonlocal diffusion using **substructuring-based DD frameworks** and **FETI-inspired methods** for efficient and scalable solvers.

The Physics Matrix: Local vs. Nonlocal

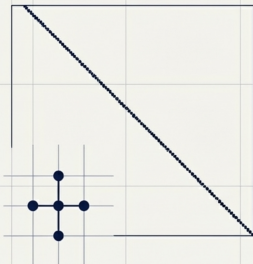
Nonlocal models capture **multiscale** phenomena, fracture propagation, and anomalous behavior that classical PDEs fail to describe.

	Classical PDEs (Local)	Nonlocal Models
Interaction Range	Infinitesimal 	Horizon δ 
Interface Definition	Lower-Dimensional Manifold 	Finite Volume 
Admissible Solutions	Continuous Functions 	Discontinuous / Jump Solutions 

The Computational Tax of Action at a Distance

Core Friction: Nonlocal interactions destroy the sparsity of traditional FEM stiffness matrices and introduce severe **ill-conditioning**.

Classical PDE Formulation



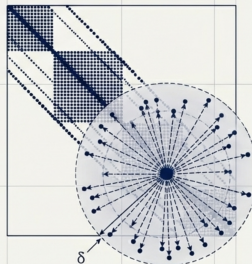
Sparse interactions. Well-conditioned.



Condition Number $\propto f(h, \delta)$

Scales adversely with both grid size (h) and horizon parameter (δ).

Nonlocal Formulation



Dense interactions. Highly coupled.

Weak Formulation of a Nonlocal Diffusion Problem

Goal: Derive a variational formulation using a nonlocal Green's identity.

Bilinear Form and Linear Functional

$$\mathcal{A}(u, v) = \int_{\Omega \cup \Gamma_D} \int_{\Omega \cup \Gamma_D} (u(\mathbf{y}) - u(\mathbf{x}))(v(\mathbf{y}) - v(\mathbf{x}))\gamma(\mathbf{x}, \mathbf{y}) \, d\mathbf{y} \, d\mathbf{x}$$

$$\mathcal{F}(v) = \int_{\Omega} v(\mathbf{x})f(\mathbf{x}) \, d\mathbf{x}$$

Function Spaces

- $W(\Omega \cup \Gamma_D)$: energy space
- $W^0(\Omega, \Gamma_D)$: test functions ($v = 0$ on Γ_D)
- $W^t(\Gamma_D)$: trace space for volume constraints

Variational Problem

Find $u \in W(\Omega \cup \Gamma_D)$ such that

$$\mathcal{A}(u, v) = \mathcal{F}(v) \quad \forall v \in W^0(\Omega, \Gamma_D), \quad u = g \text{ on } \Gamma_D$$

Energy Principle

$$\mathcal{E}(v) = \frac{1}{2}\mathcal{A}(v, v) - \mathcal{F}(v), \quad u = \arg \min \mathcal{E}$$

Notes: Well-posed if $|\Gamma_D| > 0$; singular kernels require principal value integrals.

Well-Posedness of the Nonlocal Problem

Goal: Ensure the problem admits a unique and stable solution.

1. Single-Domain Problem

- Well-posed if the Dirichlet interaction region Γ_D has non-zero measure.
- The solution u is the **unique minimizer** of the energy functional

$$\mathcal{E}(v) = \frac{1}{2}\mathcal{A}(v, v) - \mathcal{F}(v)$$

in the energy space $W(\Omega \cup \Gamma_D)$.

2. Discrete FEM Problem

- The linear system

$$\mathbb{A}\mathbf{u}_h = \mathbf{f}_h - \mathbb{B}\mathbf{g}_h$$

is well posed.

- The stiffness matrix $\mathbb{A}$ is **symmetric positive definite** when $\Gamma_D \neq \emptyset$.

Finite Element Discretization Details (Part 1)

- **FE Grid ($\mathcal{T}_h$):** Partition the domain $\hat{\Omega} \cup \Gamma_D \cup \Gamma_N$ into non-overlapping elements.
 - No hanging nodes; vertices at domain corners.
 - The triangulation respects subdomain boundaries; elements do not cross interfaces.
 - Curved boundaries approximated via straight-sided segments or isoparametric elements.

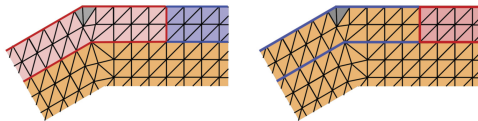


Fig. 4. Left: A zoom in of a portion of a triangular finite element grid that respects the common boundaries of the three subregions $\hat{\Omega}$ (in orange), Γ_h^D (in pink + gray), and Γ_h^N (in blue). Right: A triangular finite element grid with $\Omega_h = \hat{\Omega} \cup \Gamma_h^N$ (in orange + gray) and Γ_h^D (in pink) where Γ_h^N is bounded by the blue lines. For both cases, the two gray triangles define the geometric approximation of curved boundaries induced by the use of ℓ^2 -balls in the construction of interaction regions; specifically, those triangles are of the type depicted in gray in both plots.

Figure: Triangular FE grid showing the interior domain $\hat{\Omega}$ (orange), Dirichlet region Γ_h^D (pink), and Neumann region Γ_h^N (blue).

Finite Element Discretization Details (Part 2)

- The triangulation must respect the boundaries of the regions $\hat{\Omega}$, Γ_D , and Γ_N , so that no finite element crosses these interfaces.
- **Approximate Interaction Balls** $B_{h,\delta}$: Used instead of exact δ -balls to avoid cut elements and reduce quadrature errors.
 - Defines the discrete bilinear form $\mathcal{A}_h$ using only whole elements.

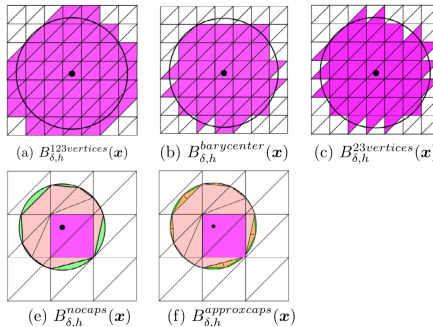


Fig. 5. Five approximations of the exact ball $B_{\delta,2}(x)$. See [41] for further details on construction and error estimates.

Figure: Triangular FE grid showing the interior domain $\hat{\Omega}$ (orange), Dirichlet region Γ_h^D (pink), and Neumann region Γ_h^N (blue).

(a), (b), (c): Whole-element approximations using only existing FE triangles, avoiding cut elements and simplifying numerical integration.

(e), (f): Refined approximations where sub-triangles are added near the boundary, providing a better representation of the ℓ_2 interaction ball and improving quadrature accuracy.

Finite Element Discretization Details (Part 3)

- **Discrete Variational Problem:**

$$\mathcal{A}_h(u_h, v_h) = \mathcal{F}(v_h) \quad \forall v_h \in W_{h,0}, \quad u_h = g_h \text{ on Dirichlet region } \Gamma_D$$

- Degrees of freedom split into $\mathcal{I}$ (interior + Neumann) and $\mathcal{B}$ (Dirichlet nodes).
 - Matrix form: $\mathbb{A}\mathbf{u}_h = \mathbf{f}_h - \mathbb{B}\mathbf{g}_h$.
 - $\mathbb{A}$ is symmetric and positive definite if $|\Gamma_D| > 0$.
 - Non-homogeneous Dirichlet conditions included as a correction to the RHS vector.
-
- *Note:* Using approximate interaction balls avoids slow convergence from cut elements, and these local matrices are then assembled into the global system for the FETI solver.

Floating Subdomains in Domain Decomposition

Floating Subdomains

- Some subdomains may not intersect the Dirichlet boundary.
- These are called **floating subdomains**.

Local Singularity

- For floating regions, the local bilinear form satisfies

$$\mathcal{A}_n(u_n, v_n) = 0$$

for constant u_n .

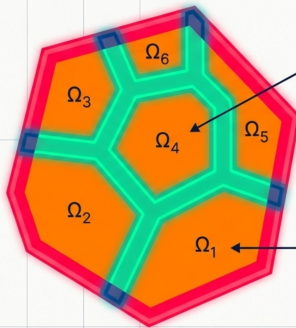
- The local stiffness matrix becomes **singular**.
- This produces a **nullspace of constant functions**.

FETI Solution

- Lagrange multipliers enforce compatibility across subdomains.
- Pseudoinverses handle the local nullspaces.

Designing the Mandatory Overlap

To calculate nonlocal energy, interfaces cannot be lines. They must expand to accommodate the full volumetric horizon.



Floating Subdomains

Internal only. Governed by pure Neumann constraints. Completely encased in mandatory δ -thick overlap ($\hat{\Gamma}_n$).

Non-Floating Subdomains

Anchor domains touching physical boundaries. Governed by Dirichlet constraints (Γ_n).

Floating Subdomains

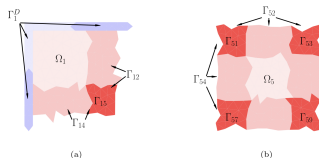


Fig. 7. (a) The overlap of $\tilde{\Omega}_1 \cup \Gamma_1^D$ with its neighboring subdomains contains Γ_{12} , Γ_{13} , and Γ_{14} . The corresponding Dirichlet domain is given by Γ_1^D (22). (b) The floating subdomain $\tilde{\Omega}_5 = \Omega_5 \cup \Gamma_5$ has an empty intersection with Γ^D . The dark red parts overlap with 3 other subdomains. See for example Γ_{31} which has overlaps with $\tilde{\Omega}_1$, $\tilde{\Omega}_2$ and $\tilde{\Omega}_4$.

Figure: Example of a floating subdomain (Ω_5).

Ω_5 is a **floating subdomain** since it does not intersect the Dirichlet region Γ_D . Interactions occur only with neighboring subdomains through overlapping regions. The local stiffness matrix is singular because constant functions lie in the nullspace.

How Do We Find $u \in W(\Omega \cup \Gamma_D)$?

Key idea: The variational problem defines an *ideal solution*. In practice, u is found through numerical discretization and efficient solvers.

1. Mathematical Layer: Energy Minimization

- Finding u is equivalent to minimizing the energy functional

$$\mathcal{E}(v) = \frac{1}{2}\mathcal{A}(v, v) - \mathcal{F}(v).$$

- A unique minimizer exists if $|\Gamma_D| > 0$.

2. Numerical Layer: Finite Element Discretization

- Replace the infinite-dimensional space by a finite space W_h
- Approximate u using nodal basis functions
- Obtain a linear system

$$\mathbb{A}\mathbf{u}_h = \mathbf{f}_h - \mathbb{B}\mathbf{g}_h$$

3. Computational Layer: FETI Solver

- Decompose the domain into subdomains
- Enforce consistency using Lagrange multipliers
- Solve iteratively in parallel (projected gradient method)

Takeaway: We compute u by discretizing the energy problem and solving the resulting matrix system using a parallel FETI algorithm.

FETI Formulation for Nonlocal FEM

Key Idea: Enforce continuity between subdomains using **Lagrange multipliers**.
Saddle Point System

$$Su_h + B^\top \lambda = f_h$$

$$Bu_h = 0$$

- u_h : discrete displacement coefficients
- λ : Lagrange multipliers enforcing interface constraints
- S : block diagonal matrix of subdomain Schur complements
- B : constraint matrix with entries $\{0, 1, -1\}$

Handling Floating Subdomains

For singular blocks, use Moore–Penrose pseudoinverse:

$$u_h = S^+(f_h - B^\top \lambda) - Z\alpha$$

- S^+ : generalized inverse
- Z : nullspace matrix (rigid body modes)
- α : nullspace coefficients

Dual System in FETI

Dual Problem

Instead of solving for u_h , FETI solves for the multipliers λ .

$$PF\lambda = Pd$$

subject to

$$G^T \lambda = e$$

Projection Operator

$$P = I - G(G^T G)^{-1} G^T$$

- Ensures compatibility with subdomain nullspaces
- Enables stable iterative solution

Solver

Projected Preconditioned Conjugate Gradient (PCG) with scaled Dirichlet preconditioner.

Computational Layer: One-Level FETI Algorithm

Goal: Solve the discrete nonlocal system efficiently in parallel

- 1 **Domain Decomposition:** Partition Ω into overlapping subdomains accounting for horizon δ
- 2 **Local Assembly:** Assemble local stiffness matrices A_n and load vectors f_n
- 3 **Interface Constraints:** Enforce continuity using Lagrange multipliers λ
- 4 **Nullspace Handling:** Construct nullspace matrix Z for floating subdomains
- 5 **Dual Problem Formulation:** Reduce the system to a dual problem for λ
- 6 **Preconditioning:** Apply Dirichlet preconditioner based on Schur complements
- 7 **Distributed Iterative Solve:** Solve using projected preconditioned CG with communication between subdomains

Final Step: Recover the global solution from local subdomain solutions

Numerical Implementation Details (Part 1)

Goal: Translate nonlocal FEM theory into a practical, solvable system while addressing challenges from long-range interactions.

1. Mesh and Subdomain Setup:

- Partition Ω into subdomains.
- Expand each subdomain by interaction horizon δ to form overlapping "interaction regions."
- Thickness of overlap δ is dictated by the model, not mesh size h .
- **Floating subdomains** (not touching Dirichlet boundary) need special treatment due to singular local stiffness.

2. Quadrature / Kernel Integration:

- Nonlocal kernels often cut through elements.
- Use **approximate interaction balls** $B_{h,\delta}$ consisting of whole elements.
- This avoids high computational cost and reduces integration errors.

Numerical Implementation Details (Part 2)

3. Local Stiffness and Mass Matrices (A_n, f_n):

- Assemble locally for each subdomain.
- Use **generalized indicator function** ζ to prevent double-counting in overlaps: weight contributions by $1/\zeta$.

4. Boundary / Volume Constraints:

- Apply Dirichlet/Neumann conditions on regions with finite measure (strips of thickness δ) rather than only surfaces.
- These are called **volume constraints**.

5. Visualization:

- Optional figure: partitioned domain with overlapping subdomains (overlap width = δ), similar to Figures 2 or 6 in the source papers.



Fig. 6. (a) The single domain $\hat{\Omega}$ (red) and the Dirichlet domain Γ^D (blue). (b) Decomposition of $\hat{\Omega}$ into 9 subdomains. The darkness of the colors indicates overlaps. The domains $\Omega_1, \dots, \Omega_9$, defined in (22), do not touch.

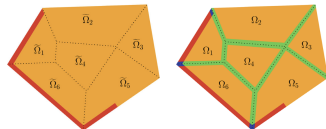


FIGURE 2 Left: a non-overlapping, covering subdivision of the domain Ω into six subdomains $\tilde{\Omega}_n$, $n = 1, \dots, N_s = 6$. Right: the corresponding nonlocal overlapping domain subdivision $\Omega_n \cup \Gamma_n$, $n = 1, \dots, N_s = 6$. The domain $\Omega \cup \Gamma$ is depicted in green and the domain $\Gamma = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_6$ is depicted in red with Γ_4 and Γ_5 being empty sets. The blue regions in Γ illustrate the overlaps between pairs of Γ_n .

Error Analysis-1

Three aspects of error are considered:

- Equivalence Error (Single vs. Multi-Domain)
- Geometric and Boundary Error
- Numerical Error and Convergence

1. Equivalence Error (Single vs. Multi-Domain)

- Domain decomposition solution is **exactly equivalent** to the single-domain problem.
- **Generalized indicator functions** (overlap counters) weight bilinear forms to avoid double-counting interactions.
- Multi-domain formulation inherits the **same discretization**, introducing **no additional approximation error**.

2. Geometric and Boundary Error

- Interaction regions defined by ℓ_2 -balls lead to **curved boundaries**.
- Controlled using **isoparametric / isogeometric approximations**.
- **Approximate interaction balls** reduce integration errors from cut elements.

3. Numerical Error and Convergence

- Weak scaling tests with $\delta = 0.008$ show decreasing L^2 error as mesh size h decreases.
- Constant kernel: error decreases from 3.47×10^{-6} ($h = 0.004$) to 4.45×10^{-8} ($h = 0.0005$).
- Observed Rate of Convergence (RoC):
 - Constant Kernel ≈ 2.0
 - Fractional Kernel ($s = 0.4$) ≈ 2.1
 - Peridynamic Kernel ≈ 2.0

Numerical Experiments: Problem Setup

Goal: Evaluate scalability and accuracy of the FETI solver.

Domain and Exact Solutions

- Computational domain: $\Omega = (0, 1)^2$
- Manufactured solution for diffusion:

$$u(\mathbf{x}) = x_1^2 x_2 + x_2^2$$

- Peridynamic model solution:

$$\mathbf{u}(\mathbf{x}) = (x_2^2, x_1^2 x_2)$$

Kernels Used in Experiments

- Constant kernel γ_c
- Fractional-type kernel γ_s
- Peridynamic kernel γ_{PD}

Baseline Solver

- Distributed Conjugate Gradient (CG), Diagonal preconditioner
- Performance measured using iteration counts and solution time

Scalability Results of the FETI Solver

Weak Scaling ($\delta = 0.008$)

- Constant kernel: CG and FETI show nearly constant iteration counts
- Fractional kernel: CG iterations grow as h^{-2s} while FETI scales better
- Peridynamic kernel: FETI provides faster solution times

Weak Scaling with Fixed Ratio ($\delta/h = 4$)

- CG solver fails to scale as the problem grows
- FETI converges in nearly constant iterations
- Significant speedups observed:
 - $7.9\times$ (constant kernel)
 - $14.6\times$ (fractional kernel)
 - $8.9\times$ (peridynamic kernel)

Strong Scaling

- Solve time decreases faster than ideal scaling
- Due to smaller subdomain problems solved with CHOLMOD

Conclusion and Future Work

Conclusion

- Developed a scalable **FETI-based domain decomposition solver** for nonlocal FEM.
- Achieves significant speedups (up to **14.6×**) compared to distributed CG.
- Demonstrates good **weak and strong scalability**.
- Multi-domain formulation is **equivalent to the single-domain problem**.

Future Work

- Extend framework to **nonlocal elasticity and peridynamics**.
- Apply to **time-dependent problems**.
- Study **nonlocal interfaces and parameter uncertainty**.
- Investigate **optimization-based DD methods** and large-scale **3D simulations**.

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